Date: 18/06/2026

Lab Practical #02:

Study of different network devices in detail.

Practical Assignment #02:

1. Give difference between below network devices.
   - Hub and Switch
   - Switch and Router
   - Router and Gateway
2. Working of below network devices:
   - Switch
   - Router
   - Gateway

Hub and Switch

No.	Hub	Switch
1	Hub is operated on Physical layer of OSI model.	While switch is operated on Data link layer of OSI Model.
2	Hub has 4/12 ports.	While switch can have 24 to 48 ports.
3	In hub, Packet filtering is not provided.	While in switch, Packet filtering is provided.
4	Hub is not an intelligent device that sends message to all ports hence it is comparatively inexpensive.	While switch is an intelligent device that sends message to selected destination so it is expensive.
5	Hub is a broadcast type transmission.	While switch is a Unicast, multicast and broadcast type transmission.

Switch and Router

No.	Switch	Router
1	The main objective of switch is to connect various devices simultaneously.	The main objective of router is to connect various networks simultaneously.
2	Switch is used by only LAN.	Router is used by LAN as well as MAN.
3	Through switch data is sent in the form of frame.	Through the router, data is sent in the form of packets.
4	Switch needs at least single network is to connect.	Router needs at least two networks to connect.
5	Switch is an expensive device than hub. but cheaper than router.	Router is a relatively much more expensive device than switch.

Router and Gateway

No.	Router	Gateway

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1	It is a hardware device that is responsible for receiving, analyzing, and forwarding data packets to other networks.	It is a device that is used for communication among networks that have a different set of protocols.
2	It supports dynamic routing.	It does not support dynamic routing.
3	The main function of a router is routing the traffic from one network to the other.	The main function of a gateway is to translate one protocol to the other.
4	A router operates on layer 3 and layer 4 of the OSI model.	A gateway operates up to layer 5 of the OSI model.
5	It is installed on router equipment in a dedicated appliance.	The gateway is installed as a virtual or physical server or a dedicated appliance.

Working of below network devices:

1. Switch

- The Switch is a network device that is used to segment the networks into different subnetworks called subnets or LAN segments. It is responsible for filtering and forwarding the packets between LAN segments based on MAC address.

- Switches have many ports, and when data arrives at any port, the destination address is examined first and some checks are also done and then it is processed to the devices. Different types of communication are supported here like unicast, multicast, and broadcast communication.

2. Router

- A Router is a networking device that forwards data packets between computer networks. One or more packet-switched networks or subnetworks can be connected using a router. By sending data packets to their intended IP addresses, it manages traffic between different networks and permits several devices to share an Internet connection. Although there are many kinds of routers, the majority of them transfer data between LANs (local area networks) and WANs (wide area networks).
- Consider a router as an air traffic controller, and consider data packets as planes flying to various airports (or networks). Each packet must be directed as quickly as possible to its destination, just as each plane has a distinct destination and travels a distinct route. A router assists in guiding data packets to their intended IP address, just like an air traffic controller ensures that aircraft reach their destinations without getting lost or experiencing significant disruptions in a route.

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3. Gateway

- A **gateway** is a network device that connects two different networks that may use different communication protocols. It acts as a translator, enabling data to flow between networks that would otherwise be incompatible.
- When a device on a local network wants to communicate with a device on another network (like accessing a website on the internet), the data is first sent to the gateway. The gateway examines the destination address and, if necessary, converts the data into a format that the other network understands.